Computer Vision — 21CSE454T

Unit 4: Vision & Motion

Topic 1: Methods for 3D Vision | SLO2

Course	Computer Vision — 21CSE454T
Unit	Unit 4: Vision & Motion
Topic	Topic 1: Methods for 3D Vision
SLO2	Implement and evaluate 3D vision algorithms.
Session Title	Evaluating 3D Vision Algorithm Outputs
Duration	10 Minutes Activity
Response Method	YES / NO cards | Thumbs 👍 / 👎 | Labelled flashcards

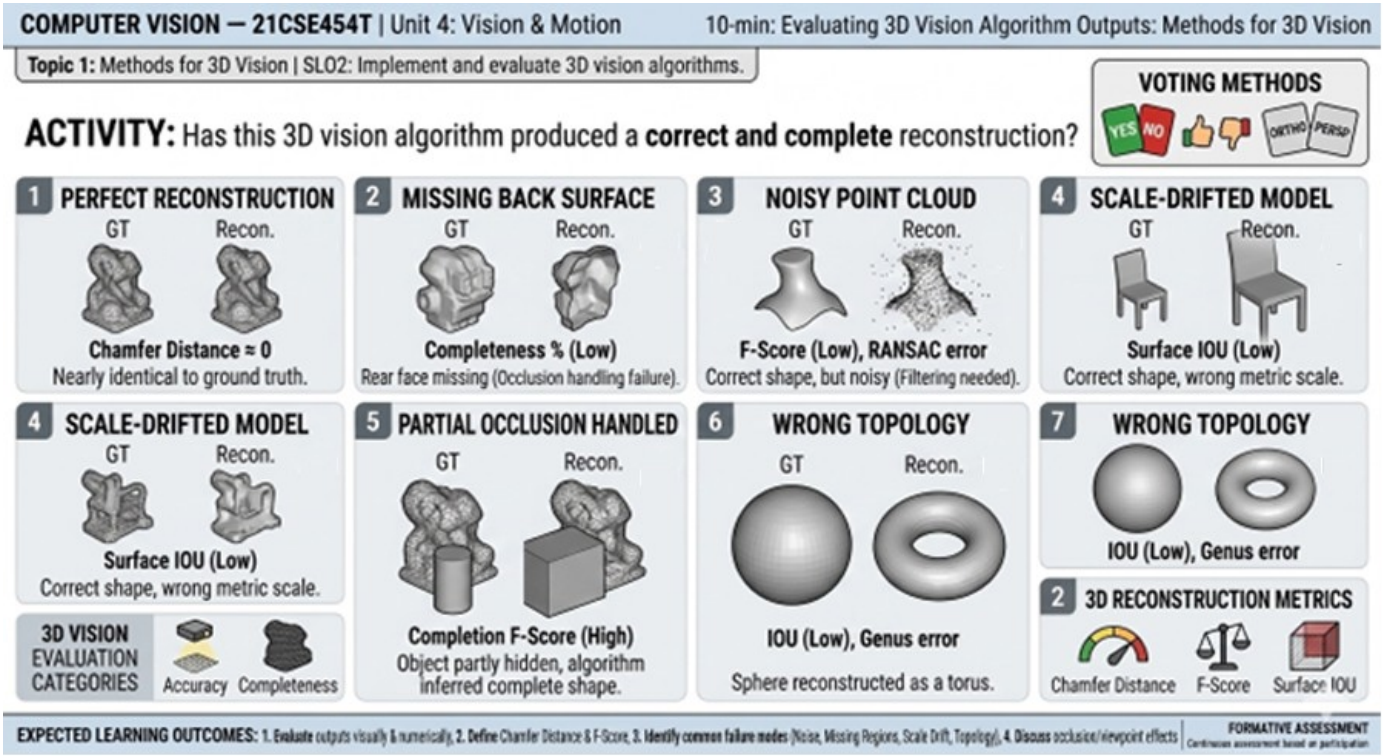
Activity Question

"Has this 3D vision algorithm produced a correct and complete reconstruction?"

Answer: YES — Perfect reconstruction; NO — Missing back surface; NO — Noisy point cloud; NO — Scale-drifted model; YES — Partial occlusion handled well; NO — Wrong topology.

Activity Overview

Students compare ground-truth 3D models against algorithm outputs (reconstructed point clouds / meshes) and vote YES/NO on reconstruction quality, developing intuition for evaluation metrics used in practice.



Instructor Instructions

1. Display 6 side-by-side pairs: ground truth 3D model vs. algorithm reconstruction output.
2. Ask: 'Is the algorithm output a correct and complete reconstruction?'
3. Students vote YES / NO for each pair.
4. Reveal the evaluation score (Chamfer distance, IOU, completeness %) after each vote.
5. Discuss what errors are present: missing regions, noise, scale drift, wrong topology.
6. Introduce common 3D reconstruction metrics: Chamfer Distance, F-Score, Surface IOU.

Step-by-Step Activity Flow

Step	Action	Details / Expected Student Response
1	**Perfect reconstruction**	Algorithm output nearly identical to ground truth. Expected: YES. Chamfer distance ≈ 0.
2	**Missing back surface**	Front face reconstructed but rear face missing. Expected: NO. Discuss occlusion handling.
3	**Noisy point cloud**	Correct shape but with many outlier points. Expected: NO. Discuss noise filtering (RANSAC, radius outlier removal).
4	**Scale-drifted model**	Correct shape but 2x too large. Expected: NO. Discuss metric scale recovery.
5	**Partial occlusion handled well**	Object partly hidden but algorithm inferred complete shape. Expected: YES. Discuss shape priors.
6	**Wrong topology**	Sphere reconstructed as a torus. Expected: NO. Discuss topological errors.

Student Response Method

After each image or scenario is displayed, students respond using one of:

- Raising YES card (green) or NO card (red)
- Showing thumbs up 👍 for YES or thumbs down 👎 for NO
- Holding up labelled category cards (e.g., ORTHO / PERSP / OBLIQUE)

The instructor acknowledges the class response, reveals the correct answer, and provides a 1–2 sentence explanation before moving to the next item.

Expected Learning Outcomes

#	Learning Outcome
1	Evaluate 3D reconstruction outputs against ground truth visually and numerically.
2	Define Chamfer Distance and F-Score as evaluation metrics for 3D vision.
3	Identify common failure modes: noise, missing regions, scale drift, topology errors.
4	Discuss how occlusion and viewpoint affect reconstruction completeness.